

Microsoft 365 Copilot Analytics

Data Onboarding Guide

Upload methods, attribute mapping, verification, and maintenance — the complete org data playbook

Document Property	Value
Section	Setup
Audience	IT Admin, Insights Admin, Data Engineer
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How This Document Fits

This is a **Setup** guide. It is the complete step-by-step guide for uploading organisational data into Microsoft 365, covering all six upload methods, attribute mapping, verification, and ongoing maintenance. Read the Data Strategy first to decide *what* to upload and *why*, then follow this guide to execute.

Section	Purpose	Who Reads
Strategy	Decide what to implement	CIO, SA, Business Leaders
Setup (you are here)	Configure and deploy	IT Admin, Insights Admin
Operate	Use dashboards and reports daily	Leaders, Analysts, Managers
Reference	Look up details and fix issues	Everyone (as needed)

Table of Contents

- 1. Determine the Current Tenant State
- 2. Upload Methods Overview
- 3. Source and Attribute Precedence

4. Upload Steps — M365 Admin Center (Primary Path)
5. Upload Steps — Viva Insights (Only When Tenant-Enabled)
6. Attribute Mapping
7. Post-Upload Verification Checklist
8. Re-Upload & Maintenance
9. Data Freshness Monitoring
10. Key Rules
11. Common Upload Errors
12. What to Read Next

1. Determine the Current Tenant State

Do not infer the supported upload path from a calendar date or from one missing tile. The migration is tenant-specific, and current Microsoft Learn pages conflict about whether Viva Insights web-app add/edit uploads remain supported.

1. Read the latest tenant-specific **Message Center** post for organizational-data migration and record its post ID and date.
2. Open `admin.microsoft.com` → **Setup** → **Migration and imports**. Record whether **Organizational Data in Microsoft 365** and its **Get started** or **New import** controls are available.
3. Open the Viva Insights web app. Record whether organizational-data **add/edit**, **delete**, and **Replace All** controls are present; these capabilities can differ.
4. Review existing connections before creating another source.
5. If Message Center and the tenant UI agree, use that path. If they conflict, or neither surface offers add/edit, stop and escalate with screenshots, roles, tenant cloud, Message Center evidence, and connection status.

Official Microsoft Sources Conflict

Manage settings for the Microsoft Copilot Dashboard, Agent Dashboard, and Viva Insights web app (Microsoft Learn, updated 13 August 2026) still recommends Viva web-app upload for Viva Insights customers. Use Microsoft Entra plus .csv files for parallel data uploads (visible date 30 June 2025; content refreshed 7 July 2026) says add/edit uploads are no longer supported there. Upload and maintain data through the Microsoft 365 admin center (13 August 2026) says tenant timing is communicated in Message Center. Follow the actual tenant state; do not silently choose one article over another.

2. Upload Methods Overview

Option	Method	Platform	Status	Refresh E
A	CSV Upload	M365 Admin Center	GA (primary path)	Incremer
B	CSV Upload	Viva Insights	Official sources conflict; tenant-specific	Full repla
C	SAP SuccessFactors Connector	M365 Admin Center	GA	Auto-syn
D	Workday Connector	M365 Admin Center	Supported up to 100,000 users	Auto-syn
E	Azure Blob Storage	M365 Admin Center	Public Preview; Worldwide only	Automat
F	API-Based Import	M365 Admin Center	Public Preview; Worldwide only	Incremer

Sources: Import organizational data from Workday (Microsoft Learn, updated 6 April 2026) supports the connector for no more than 100,000 users and documents Workday RaaS alternatives; Import organizational data using API-based import (6 April 2026) identifies API import as public preview; Supported Microsoft 365 environments for Viva Insights features (30 June 2026) lists API- and Azure blob-based organizational-data import as public preview in Worldwide and unavailable in GCC.

Supported-Environment Gate

Capability	Worldwide	GCC
CSV organizational-data import	Supported	Supported
API-based import	Public Preview	Unavailable
Azure blob-based import	Public Preview	Unavailable

The reviewed feature matrix covers Worldwide and GCC. It does not establish feature parity for GCC High or DoD; validate those environments through current Microsoft documentation, Message Center, and tenant UI before designing an ingestion path. **Source:** Supported Microsoft 365 environments for Viva Insights features (Microsoft Learn, updated 30 June 2026).

TIP: Start From Microsoft's Official CSV Template

For CSV upload (Options A and B), start from Microsoft's official organizational data .csv template on Microsoft Learn — it documents the required headers, structure, and formatting rules, which helps you avoid the most common validation errors. On the M365 Admin Center path, you download the template in-product at **Prepare organizational data → Download CSV template**.

NOTE: Decision Logic

- **Running SAP SuccessFactors?** → Use **Option C** (connector). Zero CSV.
- **Running Workday (≤100K users)?** → Use **Option D** (connector). Zero CSV.
- **Running Workday (>100K) or have a data warehouse?** → Consider Workday RaaS through **Option E** (Azure Blob) or **Option F** (API), subject to preview and cloud availability.
- **Need programmatic control or custom HRIS?** → Consider **Option F** (API), subject to preview and cloud availability.
- **Everything else?** → Use **Option A** (CSV to M365 Admin Center). Universal fallback.
- **Viva add/edit controls still explicitly enabled?** → Use **Option B** only when Message Center and tenant UI agree.

2A. Source and Attribute Precedence

Surface / attribute	Source behaviour
Organizational Data in Microsoft 365	The platform itself requires <code>Microsoft_PersonEmail</code> . Apps add their own requirements
Copilot Dashboard and Viva Insights	<code>Microsoft_PersonEmail</code> , <code>Microsoft_ManagerEmail</code> , and <code>Microsoft_OrganizationalData</code>
Microsoft 365 User Profile	Entra is the default source. Uploaded organizational data can be prioritised, and
Viva <code>ManagerId</code> and <code>Organization</code>	Parallel ingestion can select Entra as the default for either attribute while CSV/connector
Multiple upload sources	Microsoft documents merged dashboard data with the more recent upload taking

Sources: Organizational Data in Microsoft 365 — requirements and data precedence (Microsoft Learn, updated 2 December 2025); Manage settings for the Microsoft Copilot Dashboard, Agent Dashboard, and Viva Insights web app (13 August 2026); and Use Microsoft Entra plus .csv files for parallel data uploads (visible date 30 June 2025; refreshed 7 July 2026).

TIP: Bypass On-Premises AD

If your HRIS data flows through **SAP SuccessFactors → middleware → on-premises AD → Entra ID**, be aware that Viva Insights only picks up 3 attributes from Entra (`PersonId` , `ManagerId` , `Organization`). Custom fields in Entra `extensionAttributes` are **not synced**.

Solution: Use the HRIS connector to send data directly to M365 — bypassing AD entirely for analytics purposes.

3. Upload Steps — M365 Admin Center (Primary Path)

Use this path when Message Center and the live Microsoft 365 admin center show that add/edit organizational-data imports are supported for the tenant.

Step 1: Ensure Role Assignment

The person performing the upload must have the **Organizational Data Source Administrator** role assigned in Entra ID. This role manages Organizational Data in Microsoft 365 and is separate from the Insights Administrator role.

- Assign via: `admin.microsoft.com` → Users → Active Users → select user → Manage roles → search for “Organizational Data Source Administrator”

Step 2: Prepare Your CSV File

- Encoding:** UTF-8 (with or without BOM)
- File size:** Upload directly from local files up to **25 MB**. For larger CSV files, upload through SharePoint; the documented SharePoint file limit is **2 GB**.
- Attribute naming:** Use `Microsoft_*` naming convention (e.g., `Microsoft_PersonEmail`, `Microsoft_ManagerEmail`, `Microsoft_Organization`)
- No `EffectiveDate` column:** The M365 Admin Center uses a date picker in the UI, not a CSV column
- No spaces in header names:** Use underscores (e.g., `Cost_Center` not `Cost Center`)
- One row per employee:** Include every employee you want reflected in analytics

Source: Import and manage your organizational data with a .csv file — Step 3 (Microsoft Learn, updated 29 July 2026).

Step 3: Navigate to Data Connections

`admin.microsoft.com` → Setup → Migration and imports → Organizational Data in Microsoft 365 → Data Connections

Step 4: Start Import

- First upload:** Select **Get started**
- Subsequent uploads:** Select **New import**

Step 5: Select Target Apps

Choose which M365 applications will receive the uploaded data:

- **Viva Insights** — required for Copilot Dashboard and Advanced Analytics
- **Copilot Dashboard** — enables org-level filters on the dashboard
- **Microsoft 365 Profile** — populates org data in user profiles

You can select all that apply. Each app may have its own required attributes.

Step 6: Upload File and Map Reserved Attributes

- Upload your CSV file
- Map the reserved attributes: `Microsoft_PersonEmail` , `Microsoft_ManagerEmail` , `Microsoft_Organization`
- The system will auto-detect columns that match `Microsoft_*` naming

Step 7: Review Custom Attributes and Configure Access

For each custom attribute, set the access level:

Access Level	Who Can See	Use Case
Public	All selected apps and their users	Department, Location, Job Function
Confidential	Only authorised analysts (Viva Insights only)	HourlyRate, Salary Band, Performance Rating
Custom	Specific apps only	Attributes shared with Profile but not Insights

Step 8: Validate and Submit

- Review the summary and submit
- **Processing time:** A few hours for small files, up to **3 days** for large files (>100K rows)
- Monitor status in the Data Connections page: **Validating** → **Processing** → **Complete**

4. Upload Steps — Viva Insights (Only When Tenant-Enabled)

Do Not Infer This Path from a Missing Admin-Center Tile

Follow these steps only when the tenant's Message Center notice and live Viva Insights UI explicitly show add/edit CSV upload is supported. If the signals conflict, stop and escalate. Microsoft Learn currently contains conflicting statements about this path; see Section 1.

Step 1: Prepare Your CSV File

- **Encoding:** UTF-8 (with or without BOM)
- **File size:** Follow the limit shown in the live Viva upload UI; the 25 MB local / 2 GB SharePoint limits above apply to Organizational Data in Microsoft 365, not automatically to this tenant-controlled path.
- **Include `EffectiveDate` column:** The date must be **earlier** than the upload date (e.g., if uploading today, use yesterday's date or earlier)
- Use Viva Insights attribute names: `PersonId` , `ManagerId` , `Organization`

Step 2: Navigate to Data Hub

`analysis.insights.cloud.microsoft` → **Admin** → **Data hub** → **.csv upload** → **Start**

Step 3: Upload and Map Fields

- Upload your CSV file
- Map required fields: `PersonId` , `ManagerId` , `Organization` , `EffectiveDate`
- Map optional fields as needed

Step 4: Validate and Submit

- Review validation results
- **Processing time:** Up to **7 days** for large files
- Monitor status in the Data hub page

5. Attribute Mapping

This table maps Viva Insights attribute names to the Organizational Data in Microsoft 365 reserved names. Use it when preparing a file for the tenant-confirmed upload path.

Viva Insights Name	Organizational Data in Microsoft 365 Name	Required?
PersonId	Microsoft_PersonEmail	Required
ManagerId	Microsoft_ManagerEmail	Required
Organization	Microsoft_Organization	Required
EffectiveDate	N/A (date picker in UI)	Required for tenant-enabled Viva upload /
FunctionType	Microsoft_JobDiscipline	Optional
LevelDesignation	Microsoft_LevelDesignation	Optional
Location	Microsoft_CompanyOfficeLocation	Optional
HireDate	Microsoft_HireDate	Optional
HourlyRate	Microsoft_HourlyRate	Optional
Custom (e.g., BusinessUnit)	Custom (same name)	Optional

HRIS Connector and Advanced Method Documentation

For detailed setup instructions on non-CSV methods, refer to Microsoft Learn:

- SAP SuccessFactors Connector
- Workday Connector
- Azure Blob Storage
- API-Based Import

6. Post-Upload Verification Checklist

After every upload, complete this checklist to confirm data quality before relying on dashboards.

Step 1: Check Import Status

Status	Meaning	Action
Success	All rows processed. No errors.	Proceed to Step
Partial Success	Most rows processed. Some rows had warnings (e.g., unmatched manager emails).	Download impor
Failed	File rejected. No data was imported.	Check error det
Suspended	Processing was halted (usually by system or admin).	Check Data Cor

Step 2: Download Import Log

Look for rows with unmatched `Microsoft_ManagerEmail` values. These indicate managers whose email addresses don't exist in the uploaded data or in Entra ID.

Step 3: Download Manager Log

This log shows employees whose `ManagerEmail` doesn't resolve to an active user in Entra ID. These employees will appear as "orphaned" in the manager hierarchy.

Step 4: Check Attribute Coverage

Navigate to the **Data Quality** tab. Use an organization-approved acceptance threshold; this guide uses **>95% coverage for critical segmentation attributes** as an operational target, not as a Microsoft product requirement. Investigate any material drop from the pre-change baseline, because missing values can exclude people from filtered views.

Step 5: Check Manager Tree Count

The number of root nodes in the manager hierarchy should be small (typically 1–5, representing CEO/top executives). If you see hundreds or thousands, many `ManagerEmail` values are missing or incorrect.

Step 6: Spot-Check Dashboard

- Open the Copilot Dashboard and verify the **Organisation filter** shows your uploaded values (not just Entra defaults)
- Confirm new employees appear in the population count
- Check that Job Function, Location, and other optional filters are populated (if uploaded)

Step 7: Approve Connector Data (If Applicable)

If using an HRIS connector (SAP SuccessFactors or Workday) with **“Manually approve”** enabled: you have a **48-hour review window** to inspect the data before it goes live. If not approved within 48 hours, the data is automatically approved.

7. Re-Upload & Maintenance

Refresh Behaviour by Method

Method	Refresh Behaviour	Key Differences
M365 Admin Center CSV	Incremental merge	Upload only affected users. Rows and values c
Tenant-enabled Viva Insights CSV	Full replacement	Must re-upload all employees every time. Mi
HRIS connectors	Automated (weekly/monthly)	No manual uploads required. Connector sync
API-based import	Incremental or full (configurable)	Controlled via automation. Supports program

Common Maintenance Scenarios (M365 Admin Center CSV)

Scenario	What to Upload
New hire	1 row with the new employee’s data (incremental merge adds them)
Department change	1 row with updated <code>Microsoft_Organization</code> value (merge updates existing record)
Employee left	Update the authoritative employment-status/source record and the app-specific attributes rec
Historical correction	Use the Historical import option in Data Connections to submit data for a past date

Supported Update, Delete, and Historical-Correction Operations

Change	Supported operation
Update current values	Upload only affected users and include each attribute to update. Blank or omitted
Delete a string value	Use two single quotes (' ') in CSV, or three single quotes in Excel.
Delete an integer value	Use -1 for that integer attribute.
Delete an email value	Use a nonexistent email address in the tenant, such as noname@contoso.com .
Delete a date value	Use 01-01-0001 .
Correct historical Viva Insights data	Start a CSV Historical import , select the File valid as of date, and upload only the

Source: Import and manage your organizational data with a .csv file — Update or make other changes and Make retroactive updates (Microsoft Learn, updated 29 July 2026).

Recommended Cadence

- **CSV upload:** Monthly (minimum). Set a recurring calendar reminder.
- **HRIS connectors:** Weekly or monthly (configured in connector settings)
- **API-based import:** As frequently as your automation runs (daily is possible)

Do Not Use a Dummy-Row Rollback

A one-row “Replace All” upload can create an artificial hierarchy and disrupt filters, partitions, or reports. Use a corrective upload, documented deletion markers, parallel Entra restoration, or Microsoft Support.

Data-Quality Validation and Rollback Runbook

Phase	Required action and evidence
1. Baseline and backup	Preserve the exact source CSV, mapping, selected apps, access classifications, file-valid date
2. Preflight	Confirm Message Center and tenant UI path, tenant cloud, active connections, required attri
3. Validate and approve	Run import validation, review every error/warning and the attribute mapping tally, and use ma
4. Success criteria	Status reports that data is in use; processed and rejected counts reconcile to the source; no
5. Failure criteria	Failed/suspended validation, unexplained partial acceptance, population drift, new hierarchy
6. Correct or roll back	Upload corrected values for affected users; use type-specific deletion markers for values tha
7. Escalate	Open Microsoft Support when Message Center and UI conflict, processing exceeds the docu

Sources: Import and manage your organizational data with a .csv file (Microsoft Learn, updated 29 July 2026); Partitions in Viva Insights — partition-related errors (23 June 2025); and Upload and maintain data through the Microsoft 365 admin center — Support and rollback (13 August 2026).

8. Data Freshness Monitoring

Where to Check

Check	Where	What to Look For
Upload / connector status	M365 Admin Center → Data Connections	Last import date, status (Success / Partial / Failed)
Data freshness indicator	Viva Insights web app → Admin → Data hub	Freshness badge showing days since last successful sync
Attribute quality scores	M365 Admin Center → Data Quality tab	Coverage percentage per attribute (>95% target)
Dashboard spot-check	Copilot Dashboard → filters	Filter values match current org structure

Signs of Stale Data

Symptom	Likely Cause
New employees are missing from dashboards	Upload hasn't been refreshed since they were hired
Only one leader appears in Group view	Manager hierarchy data is outdated or incomplete
Old department/location values still showing in filters	Org changes not reflected in latest upload
Employee counts don't match HR system totals	Leavers not removed or new hires not added

Recommended Monitoring Cadence

When	What to Do
After every upload	Complete the Post-Upload Verification Checklist (Section 6)
Weekly (if using connectors)	Check connector health in Data Connections. Verify last sync date.
Monthly (if using CSV)	Confirm upload completion, review Message Center and Service Health, check whether counts match
Quarterly	Full audit: compare uploaded population against the HR system of record, review attribute

Sources: Tenant-specific migration communication is documented in Upload and maintain data through the Microsoft 365 admin center — Migration and timeline (Microsoft Learn, updated 13 August 2026). For current service incidents, use Microsoft 365 admin center **Health > Service health** before treating a regression as local misconfiguration.

9. Key Rules

- **Upload for every employee** — include every person you want reflected in analytics, not just Copilot-licensed users
- **No `TimeZone` column** — `TimeZone` is system-generated from Outlook settings. Including it in your CSV causes a validation error.
- **M365 Admin Center path: no `EffectiveDate` column** — the date is selected via the UI date picker
- **M365 Admin Center path: no spaces in header names** — use underscores (`Cost_Center`), not `Cost Center`)
- **Tenant-enabled Viva Insights add/edit path: include `EffectiveDate`** — follow the date rule shown in the live workflow and current Microsoft guidance
- **Incremental merge is the default** in the M365 Admin Center — you only need to upload changed or new rows. Unmentioned employees retain their existing data.
- **Data precedence is surface- and attribute-specific** — Entra is the default for Microsoft 365 User Profile, while selected uploads supply mapped Copilot Dashboard/Viva attributes; parallel ingestion can return `ManagerId` and `Organization` to Entra.
- **One import at a time per tenant** — if an import is in progress, you must wait for it to complete before starting another

10. Common Upload Errors

Error	Cause
Blank <code>Microsoft_PersonEmail</code> rows	Rows with empty email addresses
Missing required attributes for selected apps	Target app (e.g., Viva Insights) requires specific attributes
Empty data rows	Rows with all-blank values (often trailing rows from Excel)
Sending data to unlicensed apps	Selected a target app that your tenant doesn't have licenses for
Custom attributes sent to unsupported apps	Not all apps support custom attributes
Attribute name conflicts with <code>Microsoft_</code> reserved names	Custom attribute uses a name that starts with <code>Microsoft_</code>

For a comprehensive list of validation error codes and resolutions, see [Validation errors and warnings on Microsoft Learn](#).

11. What to Read Next

Ready to...	Read this
Look up the full attribute reference	Attribute Reference (Reference layer)
Troubleshoot a failed upload	Troubleshooting Guide (Reference layer)
Revisit the data strategy decision	Data Strategy (Strategy layer)
Configure roles, Viva feature access management (VFAM), and dashboards	Admin Setup Guide (Setup layer)
Validate a customer’s org data file before upload	Org Data Validation Prompt (AI-assisted va

Disclaimer

This guide is provided as a procedural reference. Always refer to Organizational Data in Microsoft 365 on Microsoft Learn for the most up-to-date information. Upload interfaces, connector status, and attribute requirements can change; Message Center and the live tenant UI control the operational path.